

Topic 12: Introduction to Probability

02-680: Essentials of Mathematics and Statistics

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Probability is the mathematical language for quantifying uncertainty

1 Basics

The sample space (Ω), sometimes called the universe, is the set of all possible *mutually exclusive* outcomes of an event.

$$\Omega_{\text{coin}} = \{\text{Heads}, \text{Tails}\}$$

$$\Omega_{\text{twocoin}} = \Omega_{\text{coin}} \times \Omega_{\text{coin}} = \{\langle \text{Heads}, \text{Heads} \rangle, \langle \text{Heads}, \text{Tails} \rangle, \langle \text{Tails}, \text{Heads} \rangle, \langle \text{Tails}, \text{Tails} \rangle\}$$

A sample outcome, or atomic element, $\omega \in \Omega$ is one of the possible things that can happen.

A set of events is a subset of the sample space with some condition. For instance if the event is that the first of two coin tosses is heads.

$$A_{\text{firstheads}} = \{\langle \text{Heads}, \text{Heads} \rangle, \langle \text{Heads}, \text{Tails} \rangle\} \subset \Omega_{\text{twocoin}}$$

We say the probability $p(A)$ of an event A is the fraction of all outcomes that A covers. So in the example above $p(A_{\text{firstheads}}) = \frac{1}{2}$.

An event space, $\mathcal{A}$, is the set of all possible events. For discrete events (such as coin flips) this can be thought of typically as the power set of Ω . In continuous spaces it is typically thought of as the Borel field of Ω (details of this are beyond the scope of the class for now).

Another way of saying this: consider non-empty Ω and $\mathcal{A}$:

- (1) $A \in \mathcal{A} \rightarrow \bar{A} \in \mathcal{A}$
- (2) $A_1, A_2, \dots \in \mathcal{A} \rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$.

1.1 Three Axioms

Axiom 1: Positive Probability. For any event A , $p(A) \geq 0$.

That is, we cannot have a negative probability. In the previous example if $A_{threetails}$ is the event that you get 3 tails when a coin is tossed twice, $p(A_{threetails}) = 0$ but this can't be negative.

Axiom 2: Total Probability. For any sample space Ω , $p(\Omega) = 1$. That is, the probability of something in the sample space happening is 1. So for the example above the probability of a single coin flip being either **Heads** or **Tails** is 1, (as defined) there are no other possible outcomes.

Axiom 3: Disjoint Events. For *disjoint* events $A_1, A_2, \dots, A_n$,

$$p(A_1 \cup A_2 \cup \dots \cup A_n) = p(A_1) + p(A_2) + \dots + p(A_n).$$

So in a single coin flip if A_{heads} and A_{tails} are the events of a single coin flip being heads and tails respectively, we can see the events are disjoint (they don't share any outcomes) so

$$p(A_{heads} \cup A_{tails}) = p(A_{heads}) + p(A_{tails}) = \frac{1}{2} + \frac{1}{2} = 1.$$

As a counter example lets define

$$A_{lastheads} = \{\langle \text{Heads}, \text{Heads} \rangle, \langle \text{Tails}, \text{Heads} \rangle\} \subset \Omega_{twocoin}.$$

We can see that

$$p(A_{firstheads} \cup A_{lastheads}) \neq p(A_{firstheads}) + p(A_{lastheads})$$

because $A_{firstheads} \cap A_{lastheads} = \{\langle \text{Heads}, \text{Heads} \rangle\} \neq \emptyset$ thus they are not disjoint.

1.2 Example Consequences of the Three Axioms

1.2.1 $p(\emptyset) = 0$

$$\Omega \cup \emptyset = \Omega.$$

Since Ω is disjoint from $\emptyset$, we know $p(\Omega \cup \emptyset) = p(\Omega) + p(\emptyset)$.

Put these two together we see that $p(\Omega) = p(\Omega) + p(\emptyset)$.

Since $p(\Omega) = 1$, then $1 = 1 + p(\emptyset)$, and thus $p(\emptyset) = 0$.

1.2.2 $p(\overline{A}) = 1 - p(A)$

We know by definition that $\overline{A} = \Omega \setminus A$.

From Axiom 3, since A and $\overline{A}$ are disjoint, we know $p(A \cup \overline{A}) = p(A) + p(\overline{A})$.

But we also know that $A \cup \overline{A} = \Omega$.

So that means $p(A) + p(\overline{A}) = p(\Omega) = 1$. With some algebra we find the original statement.

1.2.3 $0 \leq p(A) \leq 1$.

From axiom 1: $0 \leq p(A)$.

We also know $p(\overline{A}) \geq 0$ from axiom 1. (or really $-1 \times p(\overline{A}) \leq 0$)

From above we know $p(A) = 1 - p(\overline{A}) \leq 1 - 0 = 1$.

1.2.4 If $A \subseteq B$ then $p(A) \leq p(B)$.

$$\begin{aligned} p(B) &= p(A \cup (B \setminus A)) & B &= A \cup (B \setminus A) \\ &= p(A) + p(B \setminus A) & A \text{ and } B \setminus A &\text{ are disjoint (Axiom 3)} \\ &\geq p(A) & p(B \setminus A) &\geq 0 \text{ (Axiom 1)} \end{aligned}$$

2 Perspectives

For a coin toss effect, there are two methods for explaining $p(\text{Heads}) = \frac{1}{2}$:

Frequentists. You flip a coin 100 times, you get **Heads** approximately 50 of those tosses. Frequentists are very objective in the explanation of things.

Bayesian. You *believe* you will get heads 50% of the times you flip the coin. Bayesian logic is subject to its interpretation.